

Kyle Pham

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EDUCATION

University of California, Santa Barbara

Bachelor of Science in Physics

Expected June 2027

Santa Barbara, CA

Coursework: Nonlinear Dynamics, Thermodynamics, Soft Matter Physics, Fluid Mechanics, Classical Mechanics, Statistical Mechanics, Electromagnetism, Vector Calculus, Linear Algebra, Differential Equations, Physics Programming

RESEARCH EXPERIENCE

Ma Research Group & Carlini Research Group, UC Santa Barbara

Undergraduate Research Assistant – FEA Simulations

Aug 2025 – Present

Santa Barbara, CA

- Build nonlinear FEA models (COMSOL, ~360K DOF) resolving elastomeric wrinkling across 100–200% pre-strain, matching experimental wavelength and amplitude trends to within a factor of 1.4.
- Apply nonlinear continuation to solve hyperelastic bifurcations under complex multiaxial loads.
- Develop theoretical models using modified wrinkling equations to cross-validate simulations against benchtop data.
- Interface with interdisciplinary teams to translate experimental laboratory findings into computational simulation constraints.

EXPERIENCE

Cavelle Energy Technologies

Mechanical Engineering Intern

Feb 2026 – Present

Santa Barbara, CA

- Design a 5,000 PSI carbon steel flow cross in Fusion 360, adapting geometry from legacy 15,000 PSI models to streamline field handling while maintaining required safety factors.
- Create ASME-compliant manufacturing prints with GD&T and API/NPT thread callouts for CNC machining.
- Conduct static stress FEA in COMSOL to validate minimum wall thickness and resolve local stress risers.
- Present technical findings to engineering teams, using COMSOL data to justify optimizations for fabrication efficiency.

UC Santa Barbara Bicycle Shop

Bicycle Technician (Data & Metrics Team)

Jan 2024 – Mar 2026

Santa Barbara, CA

- Performed diagnostics, repairs, and preventative maintenance including drivetrain tuning and bearing overhauls.
- Managed inventory operations, component supply replenishment, and shop floor tool organization.
- Tracked and evaluated shop throughput and repair turnaround times to identify and resolve workflow bottlenecks.
- Trained campus riders on mechanical systems, component wear analysis, and maintenance techniques.

Norcal Concrete and Landscape

Project Intern

Summers, Jun 2021 – Sep 2024

Dublin, CA

- Poured reinforced concrete foundations and finished surfaces for commercial and multi-family structures.
- Laid modular pavers and soil retention assemblies engineered for high-traffic load-bearing.
- Constructed structural retaining walls, driveways, and civil walkways adhering to site specifications.
- Wired auxiliary and low-voltage control systems in compliance with regional building codes.

ENGINEERING PROJECTS

UCSB Formula SAE Electric (Gaucho Racing)

Aerodynamics Subteam Member

Mar 2025 – Nov 2025

Santa Barbara, CA

- Designed and simulated front wing and diffuser concepts to maximize downforce and minimize parasitic drag.
- Facilitated subteam trade studies to evaluate aerodynamic surface iterations and packaging tradeoffs.
- Integrated aerodynamic mounting points across chassis, suspension, and electrical systems.
- Executed trackside data acquisition and instrumentation testing to validate simulation models.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks, Autodesk Fusion 360, ASME Y14.5 GD&T, 3D Printing, DFM/DFA

Simulation & Analysis: FEA (COMSOL, ANSYS), CFD, Nonlinear/Hyperelastic Modeling, Static Stress Analysis

Programming & Computation: Python (NumPy, SciPy, Matplotlib), MATLAB, LaTeX, Git, Numerical Methods

Instrumentation & Testing: Oscilloscopes, Signal Acquisition, Mechanical Load Testing, Bearing/Drivetrain Diagnostics